

MINI PROJECT – EMPLOYEE SALARY PREDICTION ANALYTICS

Title:

Employee Salary Prediction Analytics using Python, Pandas, NumPy, Matplotlib, and Exploratory Data Analysis

Dataset: employee_salary.csv

Records: 30,000 **Columns:** 10

Departments: 6 **Salary Range:** ■25,000 to ■2,50,000

College Practical / Mini-Project Record

1. Aim

To analyze an employee salary dataset using Python and Exploratory Data Analysis techniques and to identify salary distributions, department-wise differences, experience patterns, job-level effects, and relationships between employee attributes and annual salary.

2. Abstract

The project uses the `employee_salary.csv` dataset containing 30,000 employee records and 10 columns. Python, Pandas, NumPy and Matplotlib are used for data loading, cleaning, statistical analysis and visualization. The analysis includes salary distribution, salary-range counts, department-wise average salary, job-level salary trends, experience versus salary, and correlation analysis. The project also provides a foundation for building a machine-learning model to predict annual employee salary.

3. Problem Statement

Employee salary data contains information about experience, education, department, job level, performance and other factors. It is difficult to identify salary patterns and estimate expected salary by looking at raw records alone. The problem is to analyze the supplied employee dataset and present useful salary insights through statistics, graphs and a prediction-oriented workflow.

4. Objectives

- Load and inspect the employee salary dataset.
- Identify dataset size, data types, missing values and duplicate records.
- Calculate basic salary statistics such as mean, median, minimum and maximum.
- Study the distribution of salary ranges.
- Compare average salaries across departments.
- Analyze salary differences across job levels and education.
- Study the relationship between experience and annual salary.
- Use correlation analysis to identify relationships between employee features and salary.

5. Software Requirements

Software / Library	Purpose
Python	Programming and analysis
Pandas	Data loading, cleaning and tabular analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Scikit-learn	Machine-learning salary prediction
ReportLab	Generation of this PDF report

6. Dataset Description

The analysis is based on the employee_salary.csv dataset. The dataset has 30,000 rows and 10 columns. It contains employee identifiers, demographic and professional attributes, department information, job level, performance, working hours, bonus percentage and annual salary.

Measure	Actual value
Number of rows	30,000
Number of columns	10
Departments	6
Salary range	■25,000 – ■2,50,000
Valid salary count	30,000
Duplicate records	0
Average salary	■106,837.71
Median salary	■106,483.50
Minimum salary	■25,000
Maximum salary	■205,409

Column description

Column	Description
Employee_ID	Unique employee identifier
Age	Employee age
Experience_Years	Years of professional experience
Education	Highest education category
Department	Employee department
Job_Level	Job hierarchy level
Performance_Score	Performance score
Working_Hours	Average working hours
Bonus_Percentage	Bonus percentage
Annual_Salary	Annual salary to be analyzed/predicted

7. Dataset Quality Summary

Column	Missing values	Missing %
Employee_ID	0	0.00%
Age	0	0.00%
Experience_Years	0	0.00%
Education	0	0.00%
Department	0	0.00%
Job_Level	0	0.00%
Performance_Score	0	0.00%
Working_Hours	0	0.00%
Bonus_Percentage	0	0.00%
Annual_Salary	0	0.00%

The dataset contains no missing cells in the supplied 30,000 records. No duplicate complete records were found using the complete-row duplicate check.

8. System Architecture

Stage	Process
1	Read employee_salary.csv using Pandas
2	Inspect shape, data types, missing values and duplicates
3	Prepare categorical and numerical columns for analysis
4	Calculate descriptive statistics and grouped summaries
5	Generate salary, department, job-level, experience and correlation visualizations
6	Prepare features and target for machine-learning salary prediction
7	Evaluate prediction results and report findings

Flow: Dataset → Data Loading → Data Inspection → Data Preparation → Statistical Analysis → Visualization → ML Prediction → Evaluation → Findings

9. Algorithm

Start.

Import Pandas, NumPy, Matplotlib and Scikit-learn.

Read employee_salary.csv into a Pandas DataFrame.

Display the first rows, shape, data types, missing values and duplicate count.

Calculate salary count, mean, median, minimum and maximum.

Count the records in each salary range.

Calculate average salary for each department.

Calculate average salary for each job level and education category.

Calculate the relationship between experience and annual salary.

Calculate the correlation matrix for numerical features and salary.

Generate the required graphs.

Encode categorical variables and split features and target for prediction.

Train a suitable regression model and evaluate it using MAE, RMSE and R².

Interpret the graphs and prediction results.

Stop.

10. Program 1 – Import Dataset

```
import pandas as pd
df = pd.read_csv("employee_salary.csv")
print(df.head())
```

This program imports the employee dataset using Pandas and displays the first five rows.

11. Program 2 – Dataset Information

```
print(df.info())
print(df.shape)
print(df.dtypes)
print(df.isnull().sum())
print(df.duplicated().sum())
```

This program checks the structure, size, data types, missing values and duplicate records.

12. Program 3 – Statistical Analysis

```
print(df.describe())
print("Salary Count:", df["Annual_Salary"].count())
print("Average Salary:", df["Annual_Salary"].mean())
print("Median Salary:", df["Annual_Salary"].median())
print("Minimum Salary:", df["Annual_Salary"].min())
print("Maximum Salary:", df["Annual_Salary"].max())
```

The statistical analysis gives the central tendency and range of annual salaries.

Salary statistic	Actual result
Count	30,000
Mean	■106,837.71
Median	■106,483.50
Minimum	■25,000
Maximum	■205,409
Standard deviation	■37,915.07

Salary and Range Analysis Code

```
plt.hist(df["Annual_Salary"], bins=40)
plt.xlabel("Annual Salary")
plt.ylabel("Frequency")
plt.show()
```

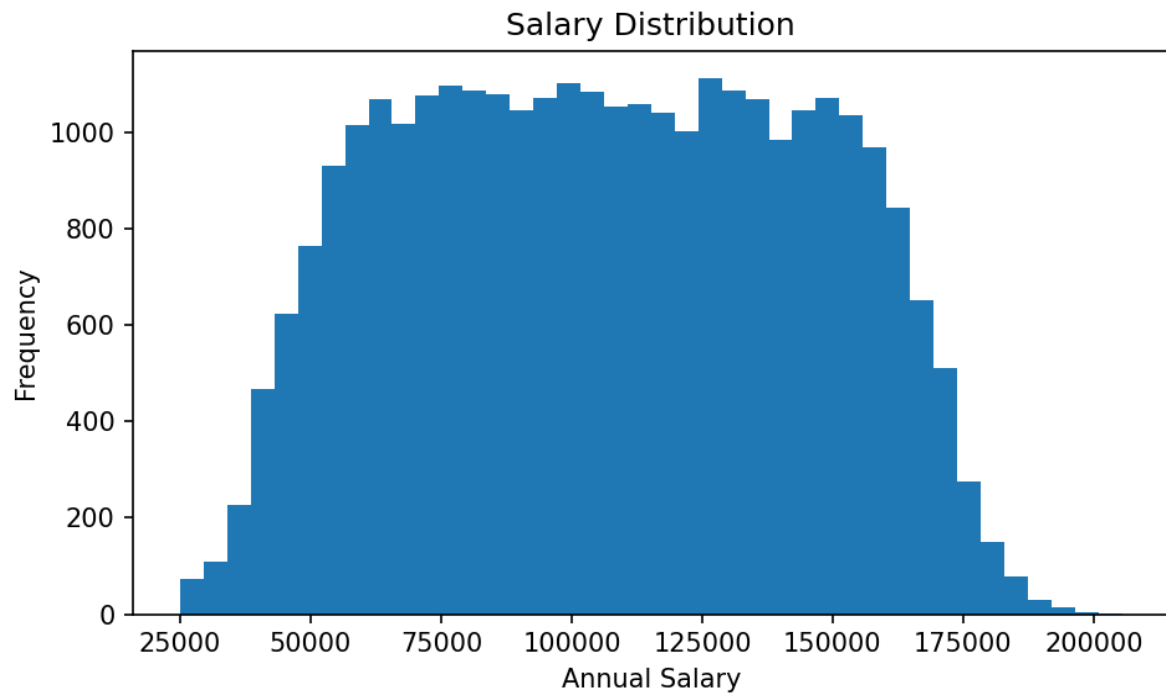
```
print(pd.cut(df["Annual_Salary"], bins=[0,50000,75000,100000,125000,150000,200000,250001]).value_counts())
```

Department and Job-Level Analysis Code

```
dept_salary = df.groupby("Department")["Annual_Salary"].mean().sort_values(ascending=False)
print(dept_salary)
level_salary = df.groupby("Job_Level")["Annual_Salary"].mean()
print(level_salary)
```

13. Analysis 1 – Salary Distribution

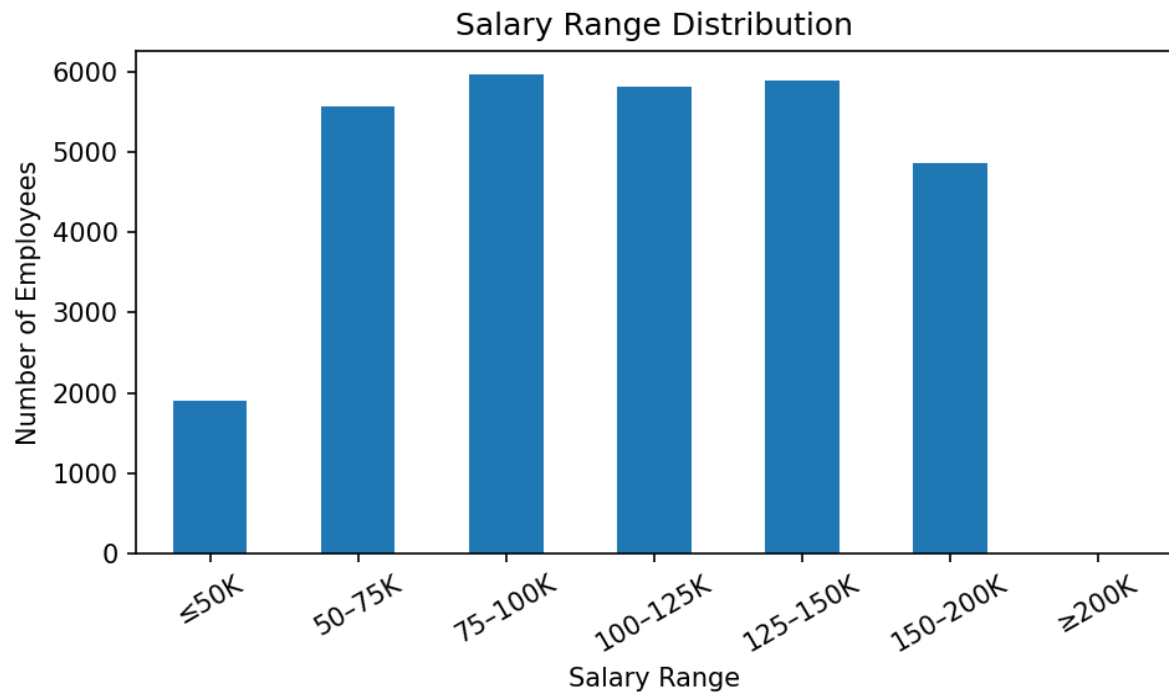
```
plt.hist(df["Annual_Salary"], bins=40)
plt.xlabel("Annual Salary")
plt.ylabel("Frequency")
plt.title("Salary Distribution")
plt.show()
```



Interpretation: The histogram shows the distribution of all 30,000 employee salary observations. The average salary is **106,837.71** and the median is **106,483.50**. The observed salary range is **25,000** to **205,409**, showing variation in employee compensation.

14. Analysis 2 – Salary Range Distribution

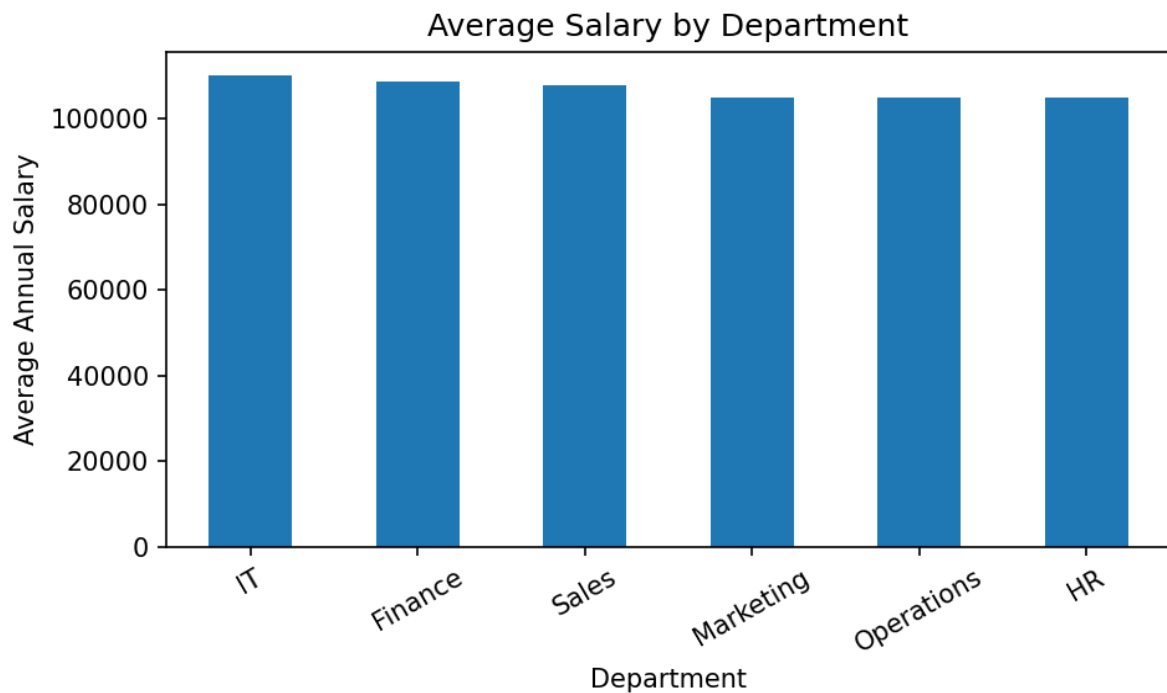
```
salary_bucket = pd.cut(df["Annual_Salary"], bins=[0,50000,75000,100000,125000,150000,200000,250001],  
labels=["≤50K","50–75K","75–100K","100–125K","125–150K","150–200K","≥200K"])  
print(salary_bucket.value_counts())
```



Interpretation: The salary-range chart groups the 30,000 employee records into practical salary bands. The most frequent salary range in this dataset is **75–100K** with 5,960 records.

15. Analysis 3 – Department-wise Salary Analysis

```
dept_salary = df.groupby("Department")["Annual_Salary"].mean().sort_values(ascending=False)
dept_salary.plot(kind="bar")
plt.xlabel("Department")
plt.ylabel("Average Annual Salary")
plt.title("Average Salary by Department")
plt.xticks(rotation=45)
plt.show()
```



Interpretation: IT has the highest average salary at ■109,991.16, while HR has the lowest average among the six departments at ■104,886.62. The chart highlights department-wise salary differences.

16. Analysis 4 – Job Level and Education Analysis

```
level_salary = df.groupby("Job_Level")["Annual_Salary"].mean()
level_salary.plot(kind="bar")
plt.xlabel("Job Level")
plt.ylabel("Average Annual Salary")
plt.title("Average Salary by Job Level")
plt.show()
```



Interpretation: Average salary generally increases across job levels in the generated dataset. Job Level 6 has the highest average salary at ■160,073.02. Education also shows differences, with PhD employees having the highest average salary among the education groups at ■117,806.82.

17. Analysis 5 – Experience vs Annual Salary

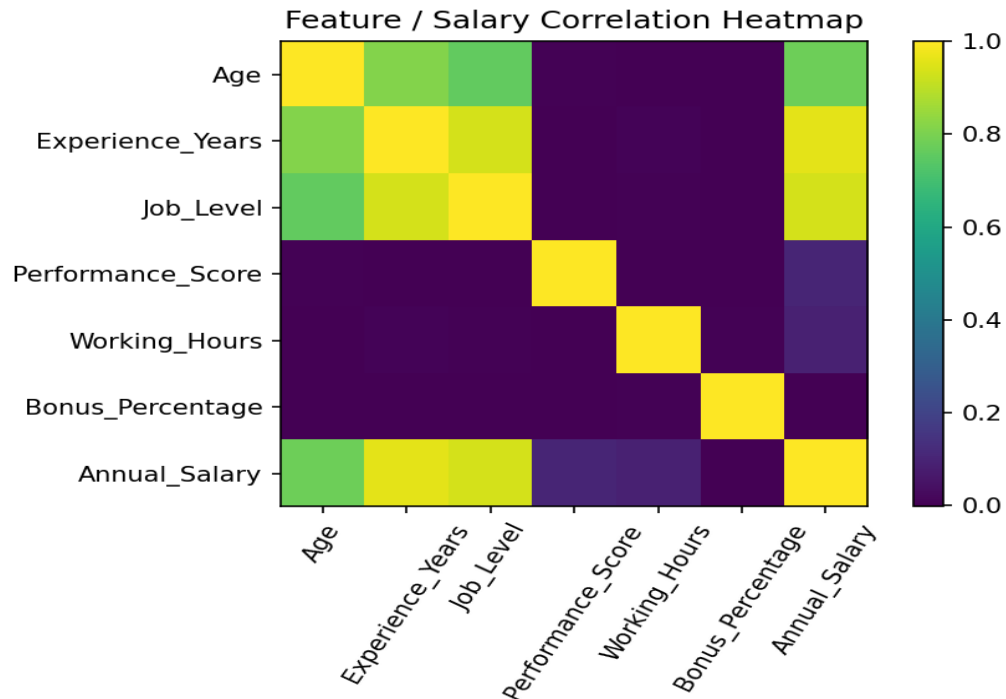
```
plt.scatter(df["Experience_Years"], df["Annual_Salary"] )  
plt.xlabel("Experience (Years)")  
plt.ylabel("Annual Salary")  
plt.title("Experience vs Annual Salary")  
plt.show()
```



Interpretation: The scatter plot compares professional experience with annual salary across 30,000 records. The Pearson correlation between experience and annual salary is approximately 0.957, indicating a positive relationship in this dataset. Employees with more experience generally tend to have higher salaries, although individual variation exists.

18. Analysis 6 – Correlation Analysis

```
features =  
["Age", "Experience_Years", "Job_Level", "Performance_Score", "Working_Hours", "Bonus_Percentage", "Annual_Salary"]  
corr = df[features].corr()  
print(corr)
```



Interpretation: The heatmap shows pairwise Pearson correlations among numerical employee variables and annual salary. The strongest absolute linear association with salary is **Experience_Years** at approximately **0.957**. Correlation describes linear association and does not by itself prove causation.

19. Analysis 7 – Salary Prediction Model

Example prediction workflow

```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

The target variable is Annual_Salary. Numerical features include Age, Experience_Years, Job_Level, Performance_Score, Working_Hours and Bonus_Percentage. Education and Department are categorical features and can be encoded using OneHotEncoder. A Random Forest Regressor can then be trained on the prepared data.

Suggested evaluation: split the data into training and testing sets, train the regression model on the training set, predict salaries for the test set, and calculate MAE, RMSE and R^2 . This step converts the descriptive analytics project into a salary prediction system.

20. Major Findings

Finding 1: The dataset contains 30,000 employee records and 10 columns.

Finding 2: The average annual salary is ■106,837.71.

Finding 3: The median annual salary is ■106,483.50.

Finding 4: The salary range is ■25,000 to ■205,409.

Finding 5: IT has the highest average salary among departments at ■109,991.16.

Finding 6: Job Level 6 has the highest average salary at ■160,073.02.

Finding 7: Experience and annual salary have a Pearson correlation of approximately 0.957.

Finding 8: The strongest numerical linear association with salary is Experience_Years, with correlation 0.957.

21. Advantages

- Simple and easy-to-understand data analysis process.
- Uses Pandas for efficient tabular data handling.
- Visualizations make salary patterns easier to understand.
- Department-wise and job-level grouping supports comparison.
- Correlation analysis helps identify related variables.
- The project can be extended directly into a machine-learning prediction system.
- Uses more than 25,000 records, supporting broader exploratory analysis.

22. Future Scope

- Build an interactive salary analytics dashboard.
- Apply Random Forest, Linear Regression, Gradient Boosting or XGBoost for salary prediction.
- Add real employee data and regularly update the dataset.
- Include location, company size, certifications and skills as additional features.
- Develop salary recommendation or benchmarking features.
- Compare multiple machine-learning models using MAE, RMSE and R^2 .
- Deploy the prediction system as a web application using Streamlit or a cloud platform.

23. Viva Questions

What is Employee Salary Analytics?

Answer: It is the analysis of employee information to understand salary patterns and factors associated with compensation.

What is the target variable?

Answer: Annual_Salary is the target variable for salary prediction.

Why is EDA important?

Answer: EDA helps understand data structure, patterns, distributions, missing values and relationships.

Which Python library is mainly used for data analysis?

Answer: Pandas is mainly used for data analysis.

Which library is used for visualization?

Answer: Matplotlib is used for creating graphs.

What is a histogram?

Answer: A histogram shows the frequency distribution of numerical values.

What is a scatter plot?

Answer: A scatter plot shows the relationship between two numerical variables.

What is correlation?

Answer: Correlation measures the strength and direction of a linear relationship.

Which feature is strongly associated with salary in this dataset?

Answer: Experience_Years, based on the numerical correlation analysis.

What is the future scope?

Answer: The project can be extended to machine-learning prediction, dashboards and salary benchmarking.

Note: Viva answers are intentionally short and based on the project dataset and analysis.

24. Result

Thus, the employee salary dataset was successfully analyzed using Python, Pandas, Matplotlib, NumPy, and Exploratory Data Analysis techniques. Various statistical analyses and visualizations were performed to identify salary distributions, department-wise differences, job-level patterns, experience relationships, and meaningful employee compensation insights. The project also establishes a clear workflow for building a machine-learning model to predict annual salary.

Key numerical summary

Item	Actual result
Rows	30,000
Columns	10
Departments	6
Salary range	■25,000 – ■205,409
Average salary	■106,837.71
Median salary	■106,483.50
Minimum salary	■25,000
Maximum salary	■205,409
Duplicate complete records	0
Highest-average department	IT (■109,991.16)
Highest-average job level	Level 6 (■160,073.02)
Experience–salary correlation	0.957
Strongest numerical salary association	Experience_Years (0.957)

End of Mini Project Record